

CYBERSYN COUCH

SPECIFICATION & BUILD PLAN



A two-seater cybernetically controlled couch inspired by Chile's 1971 Cybersyn Operations Room.

Dual independent joysticks • Switchable control modes • Shared dashboard

10 Easter eggs • MQTT/Home Assistant integration • Open hardware

Version 0.1 – March 2026

Hardware: CERN-OHL-S-2.0 | Software: MIT | Docs: CC BY-SA 4.0

Contents

1	Design Philosophy & Aesthetic
2	Physical Specifications
3	Dual Joystick System
4	Center Console Dashboard
5	Complete I/O Summary
6	Electronics Architecture
7	Software Architecture
8	Easter Egg System
9	Bill of Materials & Cost
10	Build Phases
11	Manufacturing
12	Safety & Power
A	Appendix: MQTT Topics & Config

1. Design Philosophy & Aesthetic

PRINCIPLE	DESCRIPTION
Dual Sovereignty	Each occupant has full, independent control of their joystick side
Democratic Switching	Either person can propose a control swap; both must confirm (or one can override)
Cybernetic Feedback	Every action produces multimodal feedback: visual (LEDs), auditory (tones), haptic (vibration)
No Dead Controls	Every button does something – even if it's a hidden easter egg
Accessible by Default	Large buttons, intuitive joystick, no keyboard needed (honoring the original Cybersyn ethos)
Modular & Hackable	All modules hot-swappable, all protocols open, all firmware flashable

Aesthetic Language

- **Shell:** White or off-white fiberglass, smooth compound curves
- **Upholstery:** Burnt orange (Pantone 1665 C)
- **Controls:** Backlit amber buttons, dark bezels
- **Displays:** Green-on-black OLED (retro terminal)
- **Joysticks:** Matte black aluminum, orange silicone grip
- **Accent Lighting:** Addressable RGB, default warm amber

2. Physical Specifications



Fig 2.1 – Top-down view: joystick + 4 function buttons per armrest, center dashboard between seats

DIMENSION	MEASUREMENT	NOTES
Overall width	1800 mm (71")	Includes armrests
Seating width	1400 mm (55")	Two adults comfortably
Seat depth	500 mm (20")	Standard loveseat
Seat height	420 mm (16.5")	Floor to seat surface
Backrest height	850 mm (33.5")	Floor to top
Armrest height	620 mm (24.4")	Floor to armrest top
Armrest width	200 mm (7.9")	Joystick + buttons
Armrest depth	450 mm (17.7")	Front to backrest
Center console	250 mm (9.8")	Dashboard panel
Base diameter	700 mm (27.5")	Pedestal footprint
Target weight	45–55 kg	Including all electronics

Shell Construction

Single fiberglass monocoque with five integrated zones: backrest, left armrest, left seat, right seat, right armrest. The center console is a raised section between the two seat zones.

Armrest cavities (each): Joystick mount (100×100×80mm), button panel recess (150×120×15mm), electronics bay (200×150×60mm, accessed from underside), cable channel to center console & base.

3. Dual Joystick System



Fig 3.1 – Armrest detail: Hall-effect 2-axis joystick with push button, 4× Cherry MX function buttons, RGB LED indicator strip

Joystick Hardware

SPEC	VALUE
Type	Hall-effect 2-axis analog + push button
Travel	$\pm 25^\circ$ from center, spring-return
Shaft	120mm tall, 12mm dia, anodized aluminum
Grip	35mm diameter knob, orange silicone overmold
Output	2× analog (0–3.3V per axis), 1× digital push
ADC	ADS1115 16-bit I2C – 4 channels (2 per joystick)

Mounting	M4 bolt pattern, 60mm square, drop-in from above
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5 Control Modes

MODE	LEFT JOYSTICK	RIGHT JOYSTICK	LED COLOR
1 – INDEPENDENT	Controls LEFT side	Controls RIGHT side	Amber
2 – SWAPPED	Controls RIGHT side	Controls LEFT side	Blue
3 – MERGED	X axis (pan)	Y axis (tilt)	Green
4 – PILOT/CO-PILOT	FULL control	Override only (>80%)	Red/Green
5 – CUSTOM	User-defined	User-defined	Purple

Joystick Application Mappings

The joystick outputs are abstract – what they do depends on the active application, selected via the MODE rotary encoder:

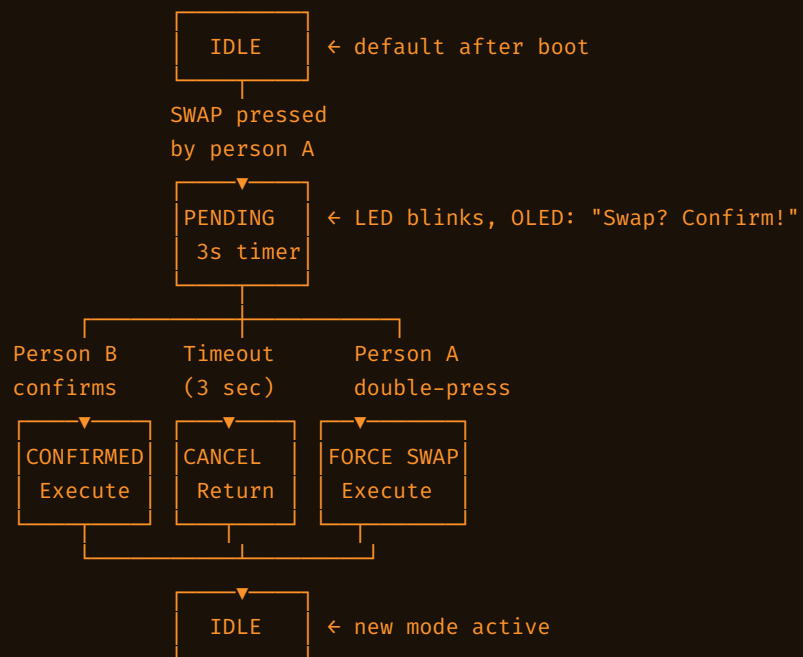
APPLICATION	X-AXIS	Y-AXIS	PUSH BUTTON
Lighting	Color temperature (warm↔cool)	Brightness (dim↔bright)	Toggle zone on/off
Music	Track skip (prev/next)	Volume (down/up)	Play/Pause
Display Navigation	Scroll horizontal	Scroll vertical	Select/Enter
Motorized (future)	Seat recline angle	Footrest extension	Lock position
Fun Mode	Spaceship steering	Throttle	Fire lasers (LED effect)
Camera/ Projector	Pan	Tilt	Shutter/toggle

Swap Mechanism

SWAP button (center dashboard):

- **Single press:** Initiates swap request – LED blinks on both armrests
- **Second person confirms** within 3 seconds: Swap executes
- **Timeout (no confirm):** Swap cancelled, brief buzz feedback
- **Double-press by same person:** Force-swap (for solo use)

- **Hold 3 seconds:** Cycle to next mode (Independent → Swapped → Merged → Pilot → Custom)

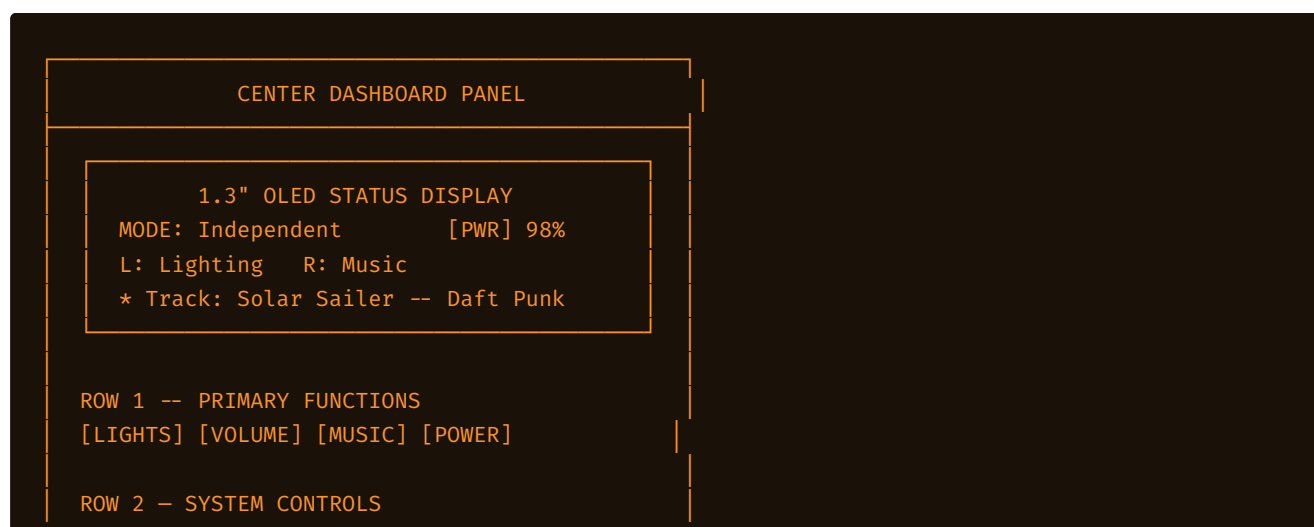


4. Center Console Dashboard



Fig 4.1 – Center console: OLED status display, primary function buttons, SWAP control, NFC reader, mode rotary encoder, easter egg row

Dashboard Layout (250mm × 180mm surface)

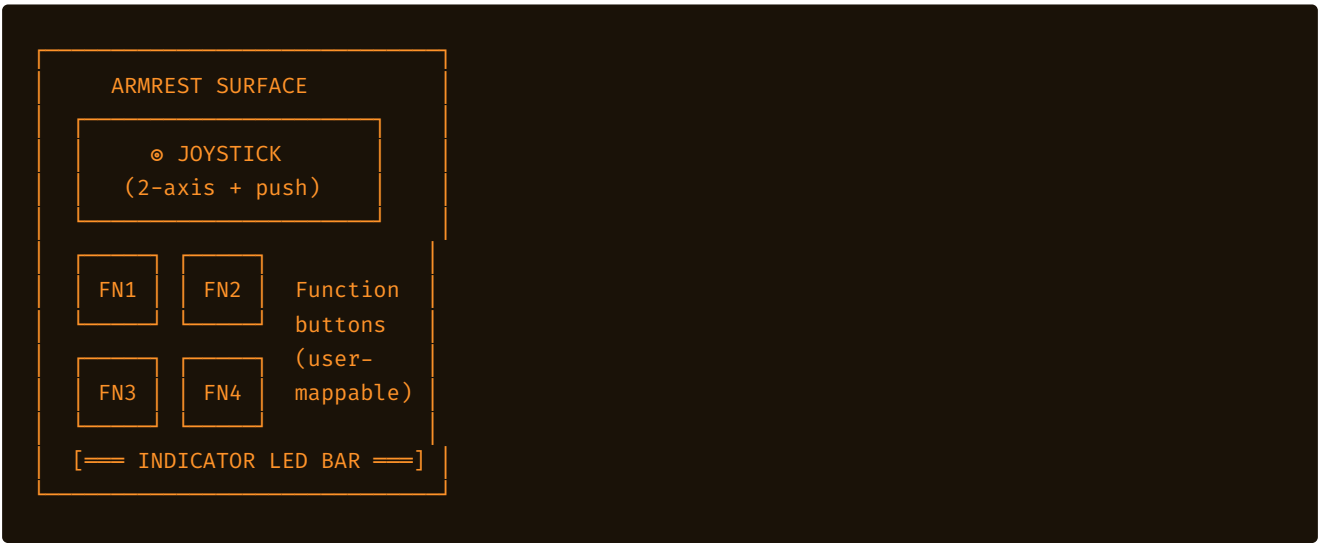




Dashboard Controls

ROW	CONTROLS	FUNCTION
Display	1.3" SH1106 OLED (128×64, green-on-black)	Mode, status, track info, zone states
Row 1: Primary	LIGHTS • VOLUME • MUSIC • POWER MODE	Master lighting toggle, rotary volume, music play/pause, system power mode cycle
Row 2: System	SWAP L↔R • NFC Reader • MODE SELECT	Joystick swap handshake, user profile loading, rotary app selector
Row 3: Easter Eggs	RANDOM • PARTY • AMBIENT • SECRET	Random Stafford Beer quote, party mode, ambient sound cycle, Konami code start

Per-Armrest Buttons (×2, mirrored layout)



BUTTON	DEFAULT MAPPING	TYPE
FN1	Zone light toggle (own side)	Cherry MX Blue, 20mm cap, RGB LED
FN2	Lighting scene cycle	Cherry MX Blue, 20mm cap, RGB LED
FN3	Playlist cycle	Cherry MX Blue, 20mm cap, RGB LED

FN4	Random easter egg (both FN4 simultaneously = Algedonic Alert!)	Cherry MX Blue, 20mm cap, RGB LED
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Button Specifications

BUTTON	TYPE	SIZE	ILLUMINATION	FEEDBACK
Joystick push	Integrated switch	N/A	Shaft ring LED	Click + haptic
FN1-FN4 (×8)	Cherry MX Blue	20×20mm cap	Individual RGB LED	Audible click + haptic
LIGHTS / MUSIC / POWER	Silicone dome, backlit	30×30mm	Amber / Green / Red LED	Haptic pulse
VOLUME / MODE	Rotary encoder + push	20–25mm knob	Ring / OLED update	Detented rotation
SWAP	Large mechanical, guarded	40×25mm	Bicolor Red/ Green LED	Heavy click + dual haptic
NFC Reader	Flush-mount circular	30mm dia	Blue ring LED	Tone on read
Easter egg row (×4)	Cherry MX, novelty caps	20×20mm	RGB programmable	Surprise!

5. Complete I/O Summary

CATEGORY	COUNT	TYPE	LOCATION
Joystick analog axes	4	Analog input (0–3.3V)	2 per armrest
Joystick push buttons	2	Digital input	1 per armrest
Function buttons	8 (4×2)	Digital input	4 per armrest
Dashboard buttons	8	Digital input	Center console
Rotary encoders	2	Quadrature + push	Center (volume + mode)
NFC reader	1	SPI	Center console
OLED displays	1–3	I2C (SH1106)	Center + optional per arm

RGB LED strips	~125 LEDs, 4 zones	WS2812B data line	Arms, base, backrest
Haptic motors	3	PWM via DRV2605L	L arm, R arm, center
Speakers	2 (stereo)	I2S or USB audio	Shell-mounted
Total GPIO-level signals	~35-40 mixed analog/digital		

LED Channel Map

INDEX RANGE	ZONE	LOCATION	DEFAULT COLOR
0-29	Left Arm	Under left armrest lip	Warm amber
30-59	Right Arm	Under right armrest lip	Warm amber
60-104	Base	Pedestal circumference	Deep orange
105-124	Backrest	Behind/under top edge	Soft orange

6. Electronics Architecture

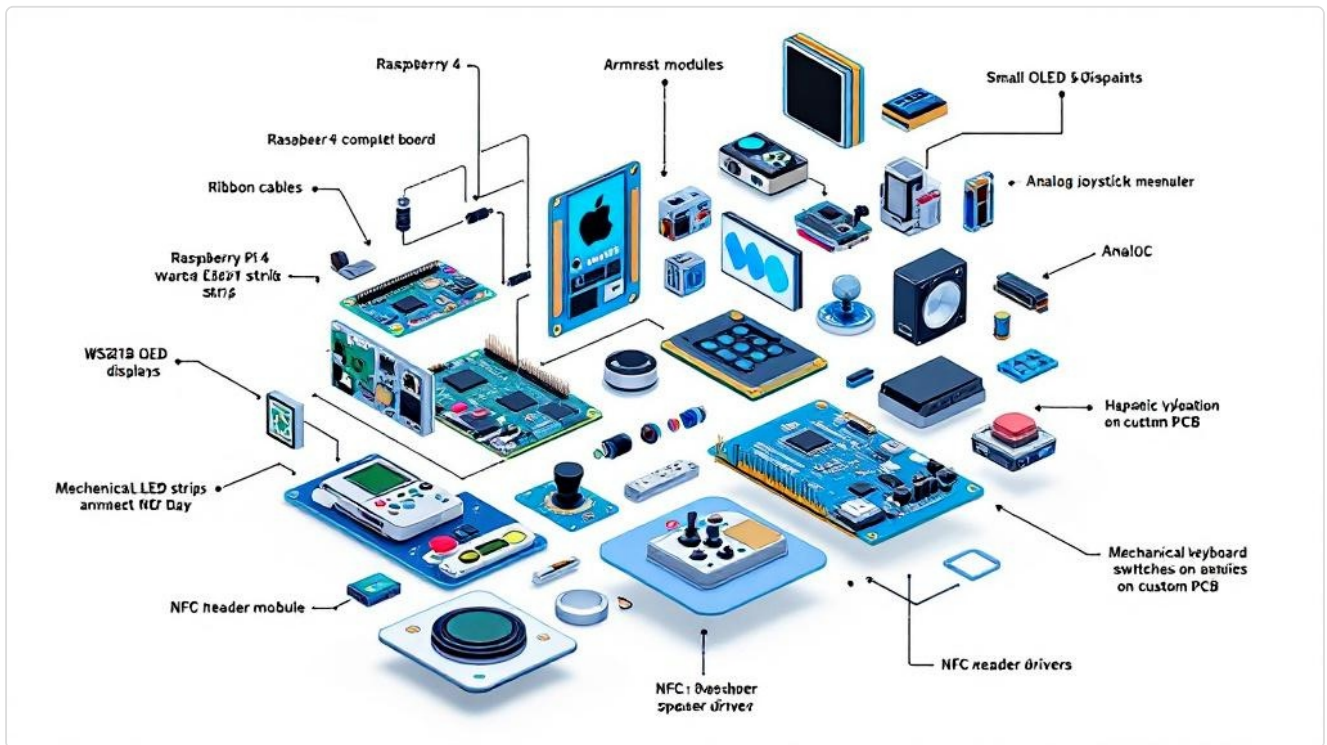


Fig 6.1 – Electronics overview: Raspberry Pi 4B brain, ADS1115 ADC, MCP23017 expanders, OLEDs, joysticks, Cherry MX PCB, haptic motors, NFC, speakers, LED strips

Core Components

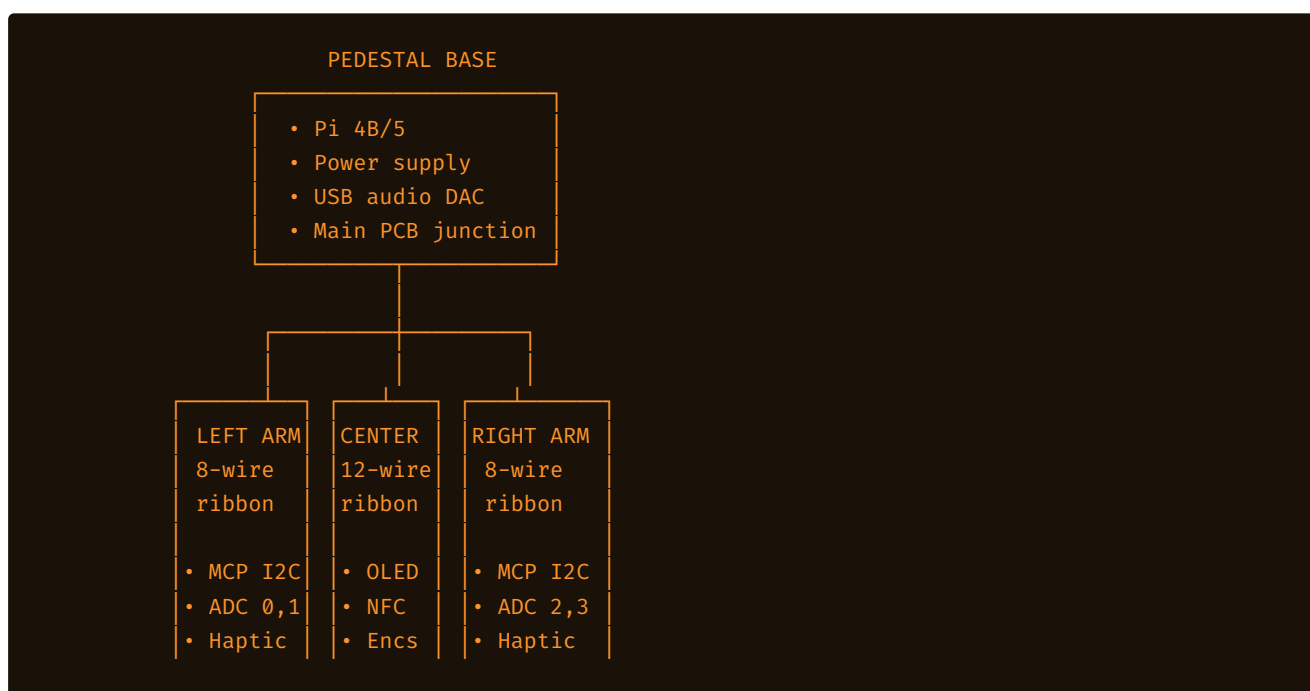
COMPONENT	PART	BUS	ROLE
Brain	Raspberry Pi 4B / 5	Host	WiFi/BT, I2C/SPI master, audio out
ADC	ADS1115 (0x48)	I2C	4-ch 16-bit for joystick analog axes
I/O Expander #1	MCP23017 (0x20)	I2C	Left arm buttons + LEDs (16 GPIO)
I/O Expander #2	MCP23017 (0x21)	I2C	Right arm + center dashboard (16 GPIO)
I2C Multiplexer	TCA9548A (0x70)	I2C	Routes duplicate-address haptic drivers
Haptic x3	DRV2605L + LRA motors	I2C (muxed)	Per-zone vibration feedback
Display(s)	SH1106 OLED 128x64	I2C	

			Status (center + optional per arm)
NFC Reader	RC522 / MFRC522	SPI	User identification / profile loading
LEDs	WS2812B strips (~125 LEDs)	GPIO data	4 zones: arms, base, backrest
Audio	USB DAC + 2× 5W speakers	USB	Music, SFX, ambient sounds

I2C Address Map

ADDRESS	DEVICE	LOCATION
0x20	MCP23017 #1	Left armrest buttons & LEDs
0x21	MCP23017 #2	Right armrest + center dashboard
0x3C	SH1106 OLED #1	Center dashboard display
0x3D	SH1106 OLED #2 (opt)	Left armrest mini display
0x3E	SH1106 OLED #3 (opt)	Right armrest mini display
0x48	ADS1115 ADC	Joystick analog inputs
0x5A–5C	DRV2605L ×3 (via mux)	Haptic motors
0x70	TCA9548A	I2C multiplexer

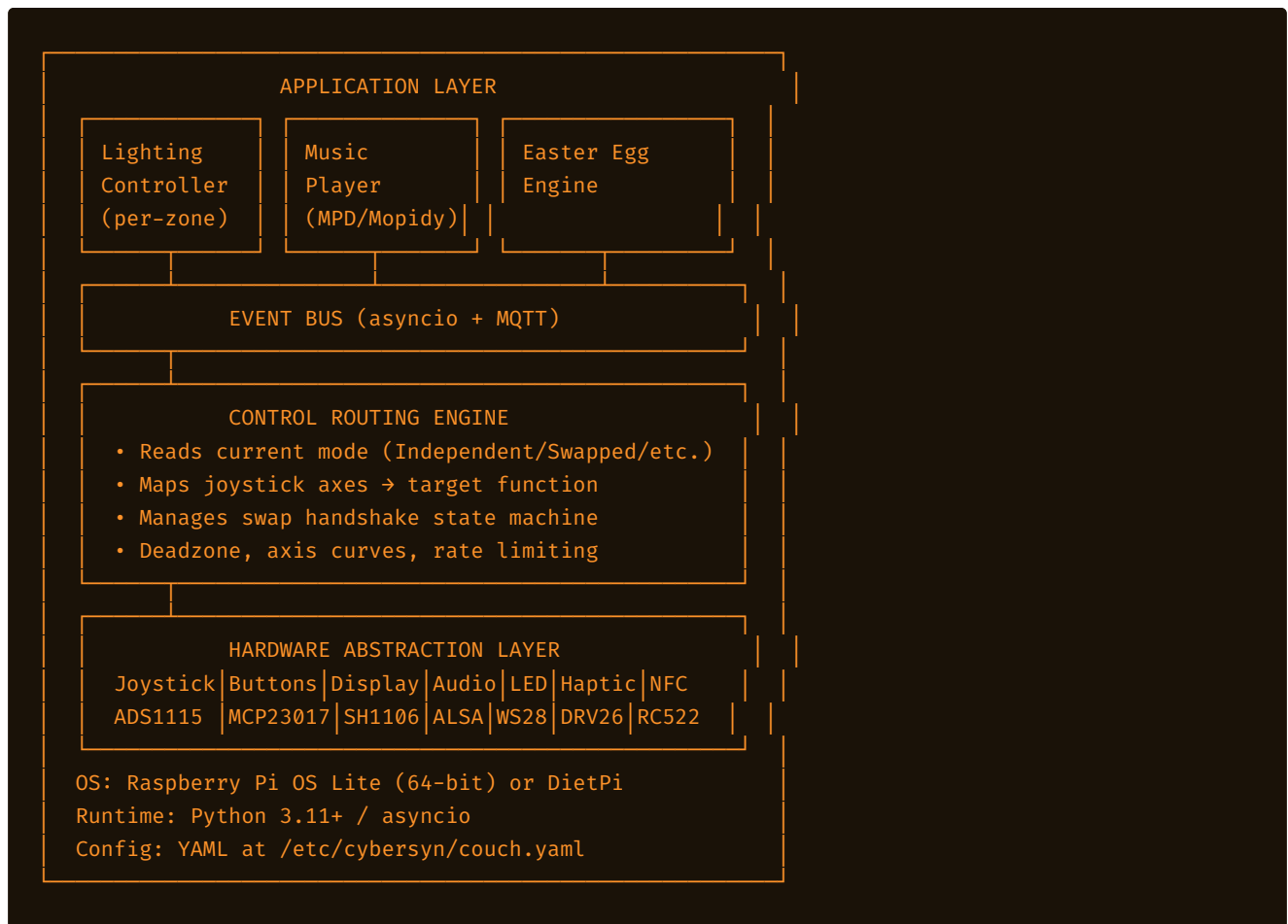
Wiring Topology





Connectors: **JST-XH** for signals, **XT30** for power. All cables in molded shell channels.

7. Software Architecture



MQTT Topic Reference

TOPIC PATTERN	PAYLOAD EXAMPLE	DIR
cybersyn/couch/{id}/joystick/{side}/axis	{"x": 0.42, "y": -0.15}	Out
cybersyn/couch/{id}/button/{loc}/{name}	{"state": "pressed", "duration_ms": 120}	Out
cybersyn/couch/{id}/mode	{"mode": "independent", "left_app": "lighting"}	Out
cybersyn/couch/{id}/swap	{"state": "pending confirmed cancelled"}	Out
cybersyn/couch/{id}/nfc	{"uid": "04:A2: ... ", "user": "operator_1"}	Out
cybersyn/couch/{id}/led/{zone}/set	{"color": [255,100,0], "effect": "steady"}	In

cybersyn/couch/{id}/audio/command	{"action": "play", "value": 50}	In
cybersyn/couch/{id}/easter_egg/ trigger	{"name": "party_mode"}	In/ Out

8. Easter Egg System



Fig 8.1 – Two operators, each with independent joystick control, warm amber LEDs – a retro-futuristic command center

10 Built-In Easter Eggs

#	NAME	TRIGGER	EFFECT	DURATION
1	Party Mode	PARTY button	Rainbow LED chase, upbeat music, haptic bass sync	Until cancelled
2	Deep Space	Hold both joystick buttons 3s	Deep blue LEDs, spacecraft ambience, star-twinkle	Until cancelled
3	Konami Code	SECRET → ↑↑↓↓←→↔ + both FN1s	Flash all LEDs, "achievement unlocked", hidden features	5 sec
4	Cybersyn Boot	Power on	Retro CRT boot on OLEDs, ascending tone, LED sweep	8 sec
5	Random Wisdom	RANDOM button	Stafford Beer quote on OLED, spoken via TTS	10 sec
6	Mood Sync	Paired NFC cards tapped simultaneously	LEDs pulse in sync between occupants	30 sec

7	Secret DJ	MODE encoder 10 full turns clockwise	DJ mode: X=crossfade, Y=filter, buttons=sample pads	Until mode change
8	Algedonic Alert	Both FN4 buttons simultaneously	All LEDs red, alarm, "ALGEDONIC ALERT" on OLED	5 sec
9	Chilean Flag	Hold LIGHTS+MUSIC+POWER together	Red, white, blue wave pattern on all LEDs	10 sec
10	Ghost in the Machine	Unoccupied for 30 minutes	Slow LED breathing, random ambient sounds at low volume	Until detected

User-Programmable: Custom easter eggs definable via YAML config – specify button sequences, timing windows, LED patterns, audio files, haptic responses, and display text.

9. Bill of Materials & Cost

Electronics BOM

ITEM	PART	QTY	UNIT	TOTAL
Main computer	Raspberry Pi 4B 4GB	1	\$55	\$55
ADC	ADS1115 16-bit I2C	1	\$12	\$12
I/O Expanders	MCP23017	2	\$4	\$8
I2C Mux	TCA9548A	1	\$6	\$6
Joysticks	Hall-effect 2-axis industrial	2	\$25	\$50
OLED Displays	1.3" SH1106 I2C	1–3	\$8	\$8–24
NFC Reader	RC522	1	\$5	\$5
Haptic Drivers	DRV2605L breakout	3	\$8	\$24
LRA Motors	10mm coin vibration	3	\$2	\$6
Mech Switches	Cherry MX Blue	12	\$1	\$12
Keycaps	Custom dye-sub PBT	12	\$3	\$36
Silicone Domes	Backlit 30mm	4	\$5	\$20
Rotary Encoders	EC11 + knobs	2	\$4	\$8
LED Strip	WS2812B 60/m	3m	\$8/m	\$24
USB Audio DAC	PCM2704	1	\$15	\$15
Speakers	2" full range 5W	2	\$8	\$16
Power Supply	USB-C PD 45W	1	\$20	\$20
Buck Converter	12V→5V 5A	1	\$8	\$8
Custom PCB	Main distribution board	1	\$15	\$15
Connectors & wire	JST-XH, ribbon, 22AWG	lot	–	\$25
Misc hardware	Standoffs, screws, heat shrink	lot	–	\$15
Electronics Subtotal				\$398–414

Structural BOM

ITEM	QTY	COST
Fiberglass cloth (6oz E-glass, 10 yd)	1	\$60
Epoxy resin (2-part laminating, 2 gal)	1	\$80
Gel coat (white, 1 gal)	1	\$45
XPS foam board 1" (plug)	4 sheets	\$120
Steel tube 4" dia, 18"	1	\$40
Steel base plate 1/4", 27" dia	1	\$80
Heavy-duty lazy susan bearing 12"	1	\$35
High-density upholstery foam 2"	2 yd	\$60
Burnt orange Ultrasuede fabric	3 yd	\$120
Sanding supplies, Bondo, primer, adhesive	lot	\$90
Structural Subtotal		\$730

Total Cost Summary

CATEGORY	COST
Electronics	\$400–415
Structural / materials	\$730
3D printed parts	\$40–60
Contingency (15%)	\$175
Grand Total (1st unit)	\$1,345–1,380
2nd unit (mold reuse)	~\$975

10. Build Phases

Estimated total: 16–23 weeks

- 0 Breadboard Proof-of-Concept** (2–3 weeks)
Wire all sensors/actuators on breadboard. Validate I2C bus (no address conflicts). Test swap state machine with two joysticks. Measure end-to-end latency (target: <20ms). Run full firmware stack with MQTT publish/subscribe.
- 1 Custom PCB Design** (2–3 weeks)
KiCad schematic & layout: Pi header, I2C/SPI buses, JST-XH connectors for each ribbon cable, screw terminals for power, test points. Fabricate at JLCPCB (2-layer), assemble and validate.
- 2 3D Printed Armrest Mockup** (2–3 weeks)
1:1 scale armrest with joystick cavity, button holes, electronics bay. Install real hardware, test ergonomics with two people. Model and print center console dashboard. Iterate at least once.
- 3 Fiberglass Shell** (4–6 weeks)
Build foam/MDF plug from CAD. Fair and finish to 400 grit. Lay up female mold, then production shell. Bond in cable channels and electronics bay walls. Sand, prime, paint white.
- 4 Pedestal Base** (2–3 weeks)
Cut and weld steel tube + base plate. Mount swivel bearing. Install Pi + PCB enclosure. Load test at 200 kg. Powder coat matte black.
- 5 Upholstery & Assembly** (2–3 weeks)
Cut foam, glue to shell, sew/install fabric. Mount all electronics, speakers, LEDs into shell. Route ribbon cables, connect everything. Mount shell to pedestal. Power-on test.
- 6 Software Integration & Tuning** (2–3 weeks)
Flash DietPi + cybersyn service. Calibrate joysticks. Configure lighting scenes and music library. Implement all 10 easter eggs. Set up Home Assistant. 8-hour soak test.
- 7 Multi-Couch Network** (future)
Couch-to-couch MQTT coordination. Voting/polling across units. Central display wall. Full 7-couch Opsroom mode.

11. Manufacturing

Shell Manufacturing Decision Matrix

FACTOR	FIBERGLASS HAND LAY-UP	ROTOMOLDING	CNC FOAM + INFUSION
Best for qty	1–6 units	7+ units	1 unit (art piece)
Surface quality	Excellent (gel coat)	Good (grainy)	Excellent
Shell weight	8–12 kg	15–20 kg	6–10 kg
Tooling cost	\$1,500–3,000	\$5,000–10,000	\$500 (CNC time)
Per-unit cost	\$300–500	\$80–150	\$400–600
Cable channels	Bond in after	Mold in	Mold in
Skill level	Intermediate	Professional	Advanced

Recommendation: Fiberglass hand lay-up for the first couch. Commission a rotomold if demand reaches 7+ units.

3D Printed Components (PETG/ABS)

PART	MATERIAL	PRINT TIME
Joystick mount plate (×2)	PETG	3h each
Button panel frame (×2)	PETG	4h each
Dashboard panel frame	ABS	5h
OLED bezel (×1–3)	PETG	1h each
Electronics bay lid (×2)	PETG	2h each
Speaker grille, NFC mount, cable clips	PLA/PETG	3.5h total

12. Safety & Power

Electrical Safety

CONCERN	MITIGATION
Fire risk	22 AWG silicone wire (200°C), 5A fuse on main
Shock risk	All 5V/3.3V DC – no mains inside couch
Short circuit	Polyfuse (resettable) per arm power feed
Heat	Ventilated cavity, passive heatsink, thermal shutdown 80°C
Pinch points	All electronics sealed behind panels

Structural Safety

CONCERN	MITIGATION
Tip-over	Weighted steel base plate, CoG analysis
Load rating	200 kg (440 lbs) – 2 adults + safety
Swivel lock	Optional friction lock on bearing
Off-gassing	Low-VOC epoxy, 7-day full cure

Power Budget

SUBSYSTEM	TYPICAL	PEAK
Raspberry Pi 4B	3.0W	6.0W
Sensors, expanders, displays	0.5W	0.9W
WS2812B LEDs (125 @ 30% avg)	7.5W	37.5W
Haptic motors (×3, pulsed)	0.1W	3.0W
Speakers (2× 5W)	2.0W	10.0W
Total	~13W	~57W

45W USB-C PD for typical use. 12V/5A (60W) supply with buck converters for full party-mode peak.

CYBERSYN COUCH

OPEN HARDWARE • OPEN SOFTWARE • OPEN FUTURE

Repository: [**gitea.jeffemmett.com/jeff/cybersyn-chair**](https://gitea.jeffemmett.com/jeff/cybersyn-chair)

Hardware: CERN Open Hardware License v2

Software: MIT License

Documentation: CC BY-SA 4.0

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